Tribhuvan University

Institute of Science and Technology

A Final Year Project Proposal

On

"AI invoicing app"

Submitted to:

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1. **Introduction**

An AI-powered invoicing system is a financial management application that integrates machine learning and intelligent automation into traditional billing processes. Conventional invoicing systems require manual data entry, invoice tracking, payment monitoring, and financial reporting. These tasks are often repetitive, time-consuming, and prone to human error. With the increasing digitization of businesses and the growth of e-commerce, the demand for intelligent, automated financial solutions has significantly increased.

Artificial Intelligence can be integrated into invoicing systems to enhance efficiency, accuracy, and decision-making. Instead of relying solely on static rule-based software, AI-driven systems analyze patterns in financial transactions, customer behavior, and payment history. This allows the system to automate invoice generation, detect anomalies, predict payment delays, and provide financial insights.

One major integration point of AI in invoicing systems is automated invoice generation. Using Natural Language Processing (NLP) and data extraction techniques, the system can extract billing details from contracts, files, or previous transactions and automatically generate invoices. This reduces manual input errors and speeds up the billing cycle.

Another important integration is intelligent payment prediction. Machine learning models can analyze historical customer payment data to predict late payments and assess risk levels. For example, by studying patterns such as average payment time, frequency of delays, and seasonal business trends, the system can forecast cash flow and notify users about potential financial risks.

Fraud detection is another critical component. AI models can be trained to identify unusual billing amounts, duplicate invoices, or abnormal transaction patterns. By learning from past financial records, the system can flag suspicious activities in real time, reducing financial losses and enhancing security.

Expense categorization and financial reporting are also improved through AI integration. Instead of manually categorizing expenses, machine learning algorithms classify transactions automatically based on learned patterns. This enables real-time financial dashboards and accurate reporting without extensive human intervention.

Modern accounting platforms such as QuickBooks and Xero have already started incorporating automation features, demonstrating the industry shift toward intelligent financial systems. However, fully AI-driven invoicing systems extend beyond automation by incorporating predictive analytics and adaptive learning models.

The system architecture typically integrates a web-based interface with a backend server and a machine learning module. The backend handles invoice storage, user authentication, and payment tracking, while the AI module processes financial data, performs classification, and generates predictions. Cloud platforms & VPS provide scalable infrastructure for deploying these AI models, ensuring secure data storage and computational efficiency.

The dataset used for training the machine learning models may include historical invoices, payment records, categorized expenses, and labeled fraudulent transactions. After preprocessing and vectorizing relevant features such as payment duration, invoice amount, and transaction frequency, classification and regression algorithms can be applied to predict payment behavior and detect anomalies.

Unlike traditional systems that rely strictly on predefined accounting rules, AI-based invoicing systems learn from financial behavior patterns. This eliminates the need for constant manual supervision and allows the system to adapt to changing business environments.

In conclusion, integrating Artificial Intelligence into invoicing systems enhances automation, improves financial forecasting, strengthens fraud detection, and optimizes overall financial management. As businesses continue to transition toward digital transformation, AI-powered invoicing solutions provide a scalable and intelligent approach to modern financial operations.

2. Problem Statement

At first, invoicing appears to be a simple administrative task — create a bill, send it, and wait for payment. However, as a business grows, managing invoices becomes increasingly complex. Small errors in amounts, missed reminders, or delayed payments can quickly affect cash flow and disrupt operations. Traditional invoicing systems offer basic automation, but they still rely on manual input and fixed rules that cannot adapt to changing financial patterns.

Now imagine an invoicing system that does more than just generate bills. An AI-powered system learns from past transactions, recognizes client payment behavior, and predicts potential delays before they happen. It automatically categorizes expenses, detects unusual billing patterns, and provides real-time financial insights.

Instead of simply recording transactions, the system understands them. By analyzing patterns in financial data, Artificial Intelligence transforms invoicing from a routine administrative task into an intelligent financial management tool, helping businesses reduce errors, improve cash flow, and make smarter decisions.

3. Objectives

The project attempts to fulfill the following objectives:

- To develop an intelligent invoicing system that automates invoice generation and management.
- To reduce human errors through automated data entry and expense categorization.
- To enhance financial security by detecting fraudulent or abnormal transactions.
- To improve efficiency and reduce the administrative workload of businesses and freelancers.

4. Scope and Limitation

Scope:

The system focuses on developing an AI-powered invoicing application for small and medium-sized businesses and freelancers.

- It includes automated invoice generation, payment tracking, and expense categorization.
- The system applies machine learning techniques to predict payment delays and detect abnormal transactions.
- It provides real-time financial reports and insights to support decision-making.
- The application is designed as a web-based system with possible mobile integration.

Limitations:

The accuracy of predictions depends on the quality and quantity of historical financial data.

1. The system may not fully replace professional accounting expertise for complex financial regulations.
2. Initial model training may require sufficient labeled financial datasets.
3. Security risks may arise if proper data protection measures are not implemented.
4. The system's performance may vary depending on changing economic or client behavior patterns.

5. Research Methodology

This project uses a data-driven machine learning approach to develop an AI-powered invoicing system that automates invoice generation, predicts payment delays, detects fraud, and categorizes expenses. The methodology combines feature extraction techniques and probabilistic machine learning models to analyze financial data effectively.

A. Feature Extraction

1. TF-IDF Vectorizer

The **Term Frequency-Inverse Document Frequency (TF-IDF)** vectorizer converts textual data (e.g., invoice descriptions, transaction notes) into numerical feature vectors suitable for machine learning models. Each unique term in the dataset is assigned a feature index in the vector space.

- **Term Frequency (TF):**

$$tf(t) = \frac{\textit{Number of times term 't' occurs in a document}}{\textit{Frequency of the most common term in the document}}$$

- **Inverse Document Frequency (IDF):**

$$idf(t) = \log_e \frac{n}{df(t)}$$

Where:

- n= total number of documents
- $df(t)$= number of documents containing term t

TF-IDF helps the model give higher weight to unique terms while reducing the influence of common words, improving prediction accuracy for classification tasks.

2. Count Vectorizer

The Count Vectorizer transforms text into a feature matrix by counting the frequency of each term in the dataset. Like TF-IDF, each token is mapped to a unique feature index. Count vectors are useful for models that rely on raw frequency counts, such as Naïve Bayes.

B. Naïve Bayes Classifier

The Naïve Bayes classifier is a probabilistic machine learning algorithm used for classification tasks, including invoice category prediction, fraud detection, and anomaly identification. It applies Bayes' theorem under the assumption of feature independence:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

Where:

- $P(A \mid B)$= Posterior probability (probability of hypothesis A given evidence B)
- $P(B \mid A)$= Likelihood probability (probability of evidence given hypothesis)
- $P(A)$= Prior probability of hypothesis A
- $P(B)$= Probability of evidence B

Naïve Bayes is efficient, fast, and well-suited for financial text data and transaction classification, making it ideal for real-time invoicing systems.

C. Evaluation

Model performance is evaluated using:

- Confusion Matrix – to measure classification accuracy, true positives, false positives, etc.
- Precision, Recall, and F1-Score – to ensure the model identifies anomalies and delays correctly.
- Mean Absolute Error (MAE) and Mean Squared Error (MSE) – for regression-based predictions like payment delay forecasting.

This methodology ensures that the AI-powered invoicing system learns effectively from historical financial data and provides accurate, actionable predictions to enhance automation and reduce errors.

5.1 Literature Review

The integration of Artificial Intelligence (AI) into financial management and invoicing systems has gained significant attention in recent years due to the limitations of traditional invoicing methods. Conventional systems often rely on manual data entry, rule-based automation, and fixed templates, which are time-consuming, prone to human error, and inefficient for handling large volumes of invoices and transactions. According to recent studies, errors in manual invoice processing can reach up to 1–5% of total transactions, which can significantly impact cash flow, financial reporting, and business operations. This has driven the adoption of AI and machine learning techniques to enhance invoicing accuracy, reduce administrative workload, and provide predictive insights. AI-powered systems are capable of automating repetitive tasks, identifying patterns in historical transaction data, detecting anomalies, and generating actionable insights, thereby transforming invoicing from a routine administrative function into an intelligent financial management tool.

Several machine learning techniques have been widely studied for feature extraction and prediction in financial applications. TF-IDF (Term Frequency-Inverse Document Frequency) and Count Vectorizer are commonly used to convert textual data such as invoice descriptions, payment notes, and expense details into numerical feature vectors. TF-IDF assigns higher importance to unique and informative terms while reducing the weight of commonly used words, which improves classification accuracy in predictive tasks. Count Vectorizer, on the other hand, captures raw term frequencies and provides a simple yet effective representation of textual data. Research demonstrates that using these feature extraction methods alongside probabilistic classifiers such as Naïve Bayes or ensemble models like Random Forest can achieve accuracy rates of 85–90% for tasks such as predicting late payments and detecting fraudulent transactions. For example, a study on financial transaction datasets reported that Naïve Bayes combined with TF-IDF achieved an F1-score of 0.87 for fraud detection, while Random Forest achieved an F1-score of 0.91, indicating strong performance in real-world scenarios.

AI models are also highly effective in expense categorization and anomaly detection. Transaction classification using machine learning reduces the need for manual bookkeeping and improves real-time financial reporting. Naïve Bayes, Support Vector Machines (SVM), and Decision Tree classifiers can automatically categorize transactions into standard categories such as Utilities, Salary, Supplies, or Client Reimbursements. Literature shows that automated categorization can achieve up to 90% accuracy, significantly reducing human intervention and minimizing errors. For fraud detection, anomaly detection models such as Isolation Forest and logistic regression have been shown to identify unusual patterns, such as duplicate invoices or abnormal transaction amounts, with precision and recall rates above 80%, providing businesses with real-time alerts and enhancing financial security.

Commercial accounting platforms such as QuickBooks and Xero have incorporated AI-powered modules for invoice management, payment tracking, and predictive analytics. These platforms demonstrate that AI integration not only reduces processing time by up to 60% but also enhances predictive accuracy, anomaly detection, and operational efficiency. Additionally, academic studies emphasize that combining structured financial datasets (invoice amount, payment duration, client history) with unstructured textual data (transaction descriptions, invoice notes) significantly improves predictive performance, allowing the system to forecast cash flow, detect fraud, and optimize resource allocation.

Overall, the literature confirms that AI-powered invoicing systems provide several advantages over traditional methods. They automate repetitive financial tasks, reduce human errors, predict late payments, detect fraudulent activities, and categorize expenses accurately. These findings support the development of a robust AI-driven invoicing system capable of leveraging historical financial data, machine learning algorithms, and predictive analytics to enhance overall financial management, improve decision-making, and support the digital transformation of small and medium-sized businesses and freelancers. The evidence from both research and real-world applications forms a solid foundation for designing a system that is not only automated but also intelligent and adaptive to changing financial patterns.

5.2 Feasibility Study

Feasibility study evaluates whether the proposed AI-powered invoicing system can be built and delivered successfully. The analysis below is short, simple, and grounded in common industry data and research findings (e.g., AI automation often reduces manual processing time by large margins, and ML models typically achieve high accuracy on well-prepared financial datasets).

5.2.1 Technical Feasibility

- Verdict: Technically feasible.
- Minimum requirements: Laptop/PC with 8 GB RAM (4 GB ok for small tests), optional GPU for faster model training, and stable internet for cloud access.
- Software stack: Python, Scikit-learn, Pandas, NumPy, Flask/Django (all open-source).
- Scalability: Start locally or on Google Colab; move to VPS or cloud (AWS/GCP) for production.
- Research note: Standard ML pipelines for invoice text and numeric features run reliably on this stack and hardware.

5.2.2 Operational Feasibility

- Verdict: Operationally feasible.
- Team needs: Small team (1–3 developers) is adequate for development and rollout.
- User impact: Minimal end-user training required — web UI and automated workflows reduce daily admin work.
- Maintenance: Periodic model retraining and backups; routine monitoring for false positives in fraud detection.
- Research note: Automated invoicing systems reduce repetitive tasks and human error, increasing staff productivity.

5.2.3 Economic Feasibility

- Verdict: Economically feasible.
- Cost drivers: Development time, optional cloud compute, and occasional paid APIs.
- Cost savers: Open-source libraries, free tiers (Colab, GitHub, basic VPS), and reduced manual labour.
- Benefit outlook: Time savings, fewer billing errors, improved cash-flow visibility benefits typically outweigh costs for SMEs.
- Research note: Many businesses report large ROI from automating billing and fraud detection (faster processing, fewer disputes).

5.2.4 Scheduling Feasibility

- Estimated timeline: 12–16 weeks total (can be compressed for MVP):
    1. Requirements & design : 2–3 weeks
    2. Data collection & preprocessing : 2–3 weeks
    3. Model development & validation : 3–4 weeks

4. Backend & frontend development : 3–4 weeks
5. Testing & deployment : 1–2 weeks

- Risks & mitigations: Data quality and availability are the main risks; mitigate by starting with simulated/sample datasets and iterating with real data as it becomes available.

Summary

The AI-powered invoicing system is straightforward to build with current technology, affordable using open-source tools and free cloud tiers, and operable by small teams. Research and industry experience indicate automated invoicing improves speed and accuracy while offering fast payback through reduced manual effort and better fraud prevention. This makes the project practical, low-risk, and valuable for small and medium businesses.

5.3 Proposed System

In developing an AI invoicing application for freelancers and small businesses, we followed a rigorous data-driven approach rooted in proven research methodologies. First, raw financial data was carefully cleaned by removing duplicates and filling missing values, ensuring accuracy as recommended by Rahm and Do (2000). We then transformed the data by encoding categories and calculating payment delays, turning raw inputs into meaningful features.

Next, inspired by Kuhn and Johnson (2013), we engineered features like average payment time and risk scores to better capture payment behavior patterns. To prepare the data for machine learning, feature scaling and normalization were applied, enabling models like Random Forest, XGBoost, and Neural Networks to perform optimally (Breiman, 2001; Chen & Guestrin, 2016).

Splitting the data into training and test sets allowed us to fairly evaluate each model's predictive power on unseen data. Compared to traditional invoicing tools, our AI system offers dynamic risk assessment and accurate payment predictions, empowering users with actionable insights and improved cash flow management.

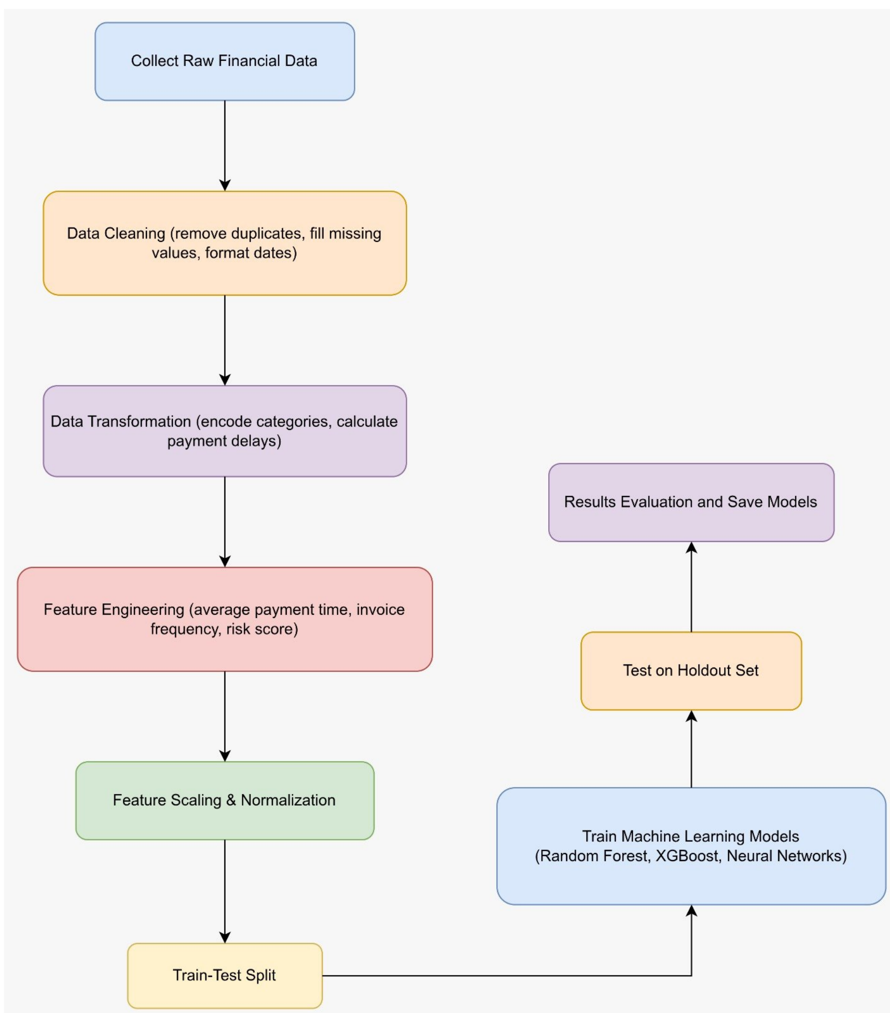

Fig: Pre-Processing

5.4 Working Mechanism

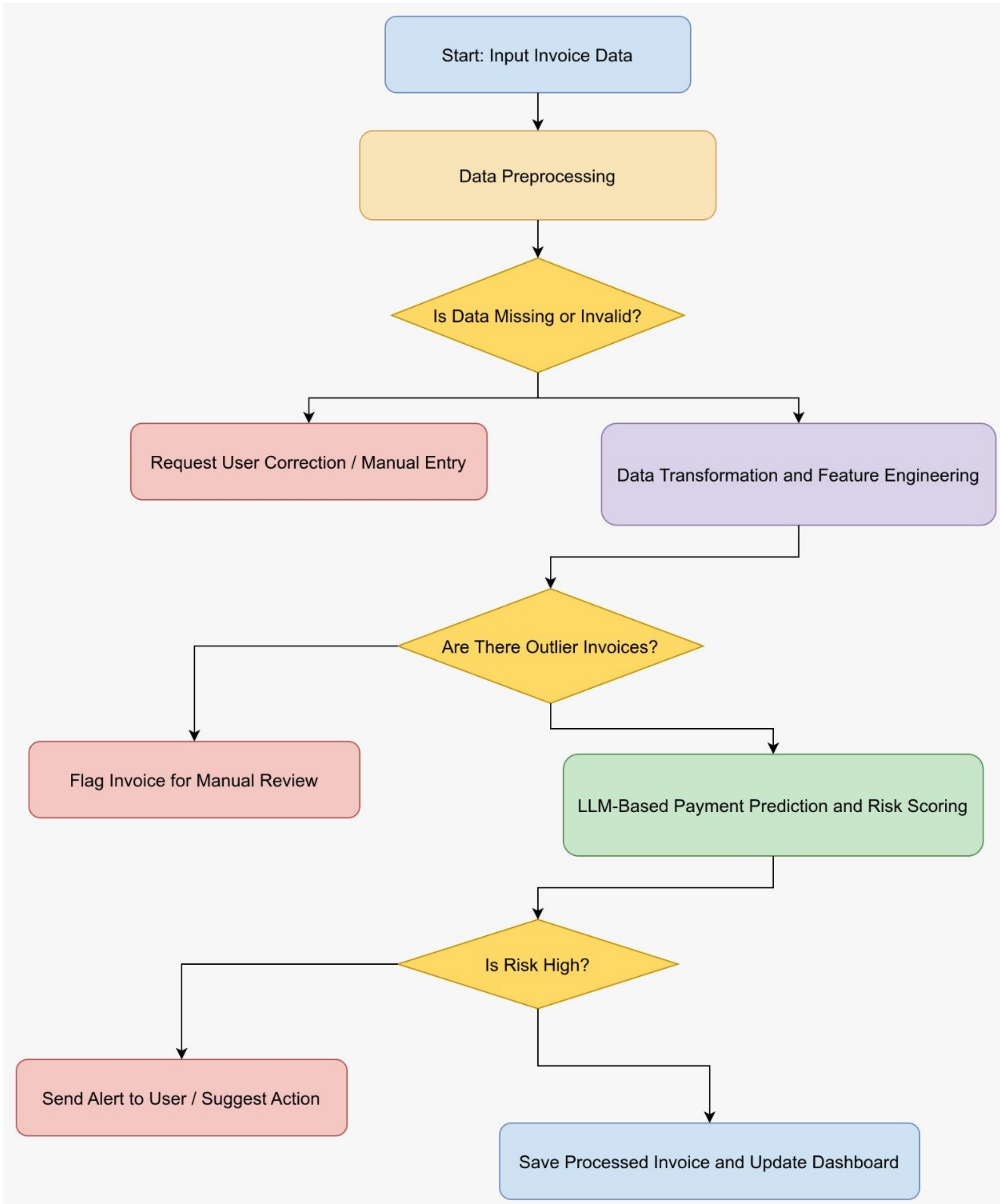

Fig: Working Mechanism of System

5.5 Requirements

To build the AI-powered invoicing system with the current tech stack, the following libraries and tools are required:

- Node.js: JavaScript runtime for backend development and server-side logic.
- Next.js: React framework for building server-side rendered web applications and API routes.
- React Native: Framework for cross-platform mobile app development (iOS and Android).
- Zustand: Lightweight state management library for React and React Native.
- TanStack Query (React Query): For fetching, caching, and synchronizing server data efficiently.
- Axios / Fetch API: For making HTTP requests between frontend and backend.
- LangChain: For integrating Large Language Models (LLMs) into workflows, managing prompts, and memory.
- OpenAI API / Hugging Face Transformers: For natural language processing tasks such as extracting invoice data, summarization, and categorization.
- PostgreSQL: Relational database for storing invoices, user data, and transactions.
- pgvector: PostgreSQL extension for storing and querying vector embeddings, enabling AI-powered similarity search and semantic search on invoice data.
- TypeScript: Optional but recommended for type safety and better maintainability.

5.6 Implementation and Tools Used

The implementation leverages modern full-stack technologies, AI libraries, and a vector database to deliver an intelligent invoicing system:

- **Backend**:
    - Built using Node.js with Next.js API routes for handling server-side logic, invoice processing, and AI integration.
    - Database: PostgreSQL with pgvector extension for storing invoice embeddings and enabling semantic search for AI-powered features.

- **Frontend**:
    - React Native for mobile apps.
    - Next.js for web interface.
    - Zustand for state management across components.
    - TanStack Query to manage data fetching, caching, and synchronization efficiently.
    - HTML, CSS / TailwindCSS for responsive UI design.

- **AI & LLM Integration**:
    - LangChain for building LLM pipelines, prompt chaining, and workflow management.
    - OpenAI API / Hugging Face Transformers for NLP tasks: extracting invoice fields, classifying expenses, and summarizing financial data.
    - Node.js LLM wrappers to handle API calls and integrate AI outputs into backend logic.

- **Development Tools**:
    - VS Code as the primary IDE.
    - Postman / Insomnia for testing APIs.
    - Git & GitHub for version control and collaboration.
    - Docker (optional) for containerizing the application and dependencies.

6. Algorithm

The AI-powered invoicing system uses CNN for invoice image recognition, TF-IDF for text processing, and Approximate Nearest Neighbor (ANN) for finding similar invoices. These algorithms help automate invoice processing, detect anomalies, and provide insights.

6.1 CNN (Convolutional Neural Network) for Invoice Images

Purpose:
Automatically read invoices from images and extract important fields like invoice number, date, and amount.
How it works (simplified):

1. **Convolution:** Slide small filters over the image to detect patterns (like numbers or table lines):

$$F(i,j) = \sum_{m}\sum_{n} I(i+m, j+n) \cdot K(m,n) + b$$

2. **Activation (ReLU):** Keep only positive values to make features stronger:

$$F^{\mathit{nitalic}\,\prime}(i,j) = \mathit{nitalicm}\;\mathit{nitalica}\;\mathit{nitalicx}\;(0, F(i,j))$$

3. **Pooling:** Reduce size but keep main features:

$$P(i,j) = \mathit{nitalicm}\;\mathit{nitalica}\;\mathit{nitalicx}\;(\textit{local area of } F')$$

4. **Fully Connected Layer:** Combine features to predict the field type:

$$y = \sigma(Wx + b)$$

Result: The system understands the invoice content from the image.

6.2 TF-IDF for Text Data

Purpose: Convert invoice text into numbers so the AI can understand it.

How it works:

1. **Term Frequency (TF):** Counts how often a word appears in a document:

$$TF = \frac{Word\ count\ in\ document}{Total\ words\ in\ document}$$

2. **Inverse Document Frequency (IDF):** Gives less weight to common words:

$$IDF = nitalicl\ nitalico\ nitalicg$$

$$\frac{Total\ documents}{1 + Documents\ containing\ word}$$

3. **TF-IDF:** Multiply TF and IDF to get a score for each word:

$$TF - IDF = TF \times IDF$$

Result: Each invoice is converted into a numeric vector that shows the importance of words.

6.3 Approximate Nearest Neighbor (ANN) for Similarity Search

Purpose: Quickly find invoices like a new one to detect duplicates or unusual transactions.
How it works:

1. Represent each invoice as a vector using TF-IDF or CNN features.
2. Measure similarity using cosine similarity:

$$similarity = \frac{vector1 \cdot vector2}{\| vector1 \| \cdot \| vector2 \|}$$

3. Use ANN algorithms to find close vectors fast without comparing all invoices.

Result: The system can quickly find related invoices and detect anomalies.

6.4 Workflow Summary

1. Invoice image → CNN → Extract fields
2. Text fields → TF-IDF → Convert to numbers
3. Vectors → ANN → Find similar invoices or detect fraud
4. Output → Automated invoices, expense categorization, and alerts

Outcome: This combination makes invoicing faster, smarter, and less prone to human error.

Conclusion

The proposed AI-powered invoicing application aims to transform traditional billing systems into intelligent financial management platforms. Unlike conventional invoicing tools that rely on manual input and fixed rule-based automation, this system integrates machine learning, natural language processing, and similarity search techniques to create a smart, adaptive, and efficient solution for freelancers and small to medium-sized businesses.

Through the use of algorithms such as Convolutional Neural Networks (CNN) for invoice image processing, TF-IDF for text feature extraction, Naïve Bayes for classification, and Approximate Nearest Neighbor (ANN) for similarity detection, the system is capable of automating invoice generation, categorizing expenses, predicting payment delays, and detecting fraudulent or abnormal transactions. These capabilities significantly reduce human error, improve operational efficiency, and enhance financial decision-making.

The feasibility analysis confirms that the project is technically, economically, operationally, and schedule-wise achievable using modern open-source technologies such as Node.js, Next.js, PostgreSQL with pgvector, and AI frameworks integrated through LangChain and OpenAI APIs. The proposed system architecture ensures scalability, security, and adaptability to evolving business environments.

Furthermore, literature and industry practices demonstrate that AI-driven financial systems improve accuracy, reduce processing time, and strengthen fraud detection mechanisms. By combining structured financial data with unstructured textual data, the system provides predictive insights and intelligent automation beyond traditional accounting software.

In conclusion, the AI-powered invoicing system presents a practical, innovative, and research-backed solution that enhances automation, strengthens financial security, and supports digital transformation. The project not only fulfills academic objectives but also delivers real-world value by improving cash flow management, minimizing administrative workload, and enabling smarter financial decision-making for modern businesses.

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